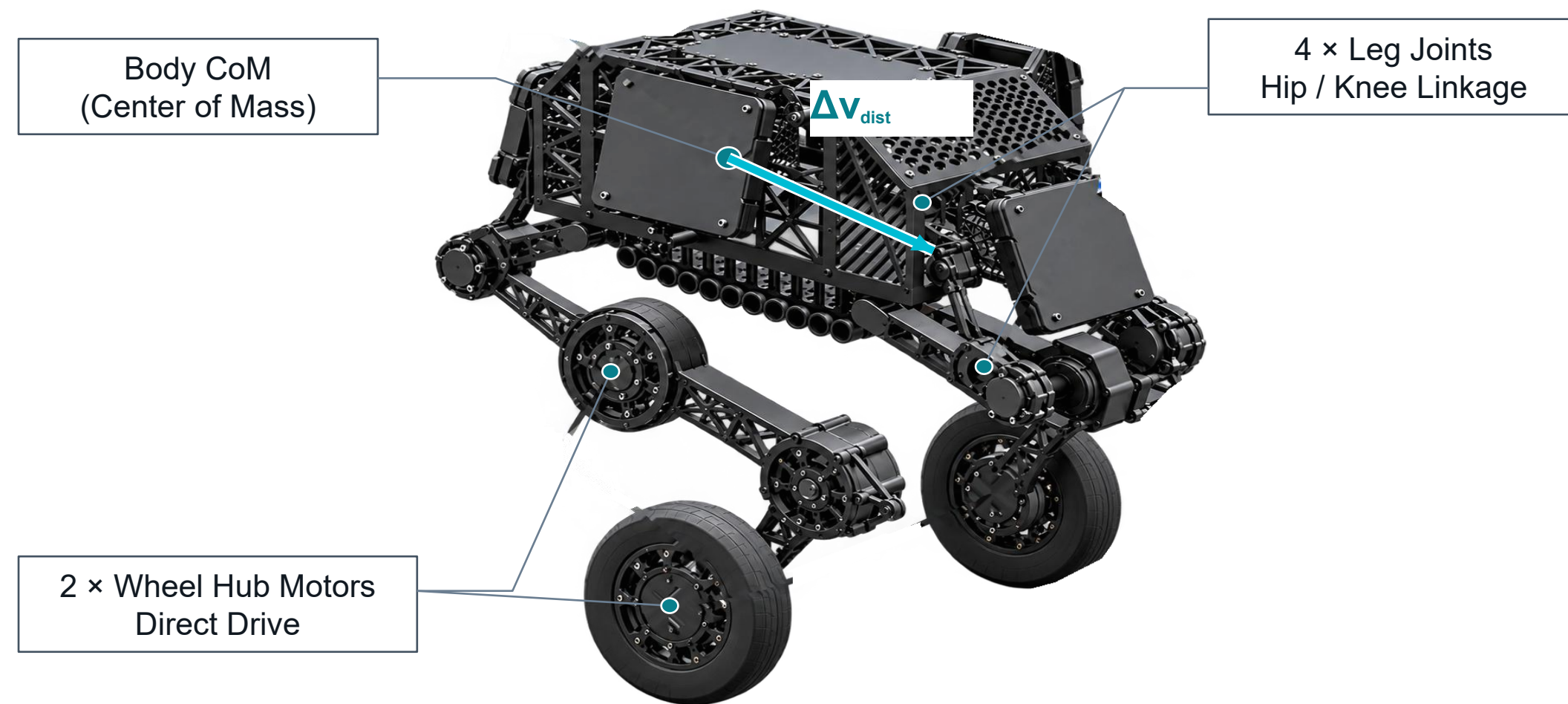


Posture-Aware Hybrid Control for a Two-Wheeled Legged Robot

From Analytical Compensation to Bounded Residual Recovery



HopperTrex Kinodynamic Model & Perturbation Profile

Key Contributions

01 Posture-aware nominal control

Model posture-induced drift; shape references. 83.68% less drift · 94.44% lower speed peaks.

02 Targeted wheel-leg residuals

12.42% shorter recovery; leg masking reduces the matched-baseline gain to 3.82%.

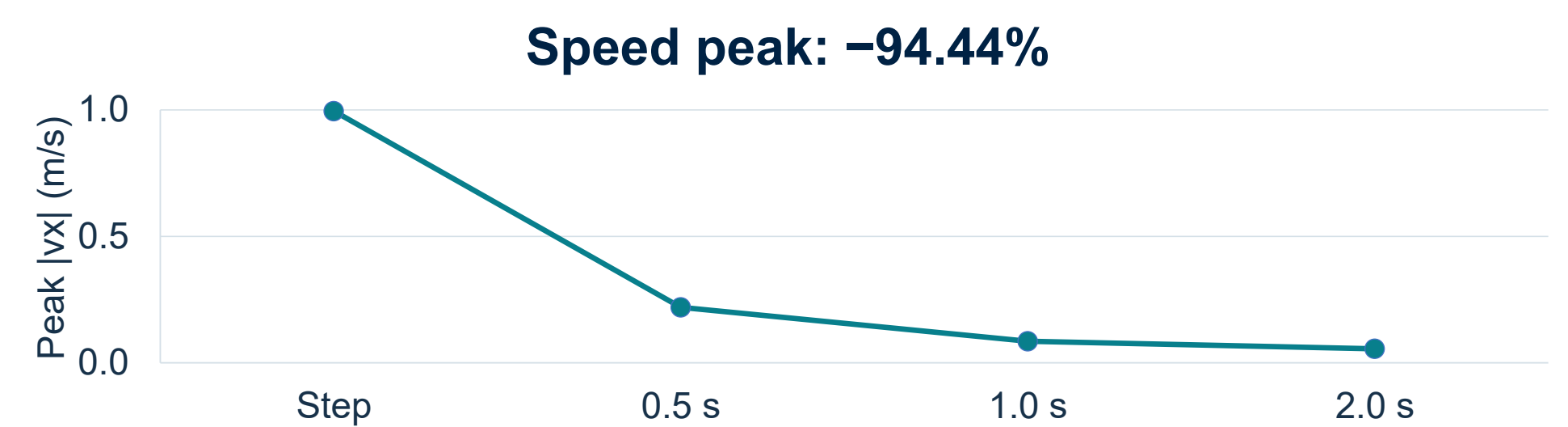
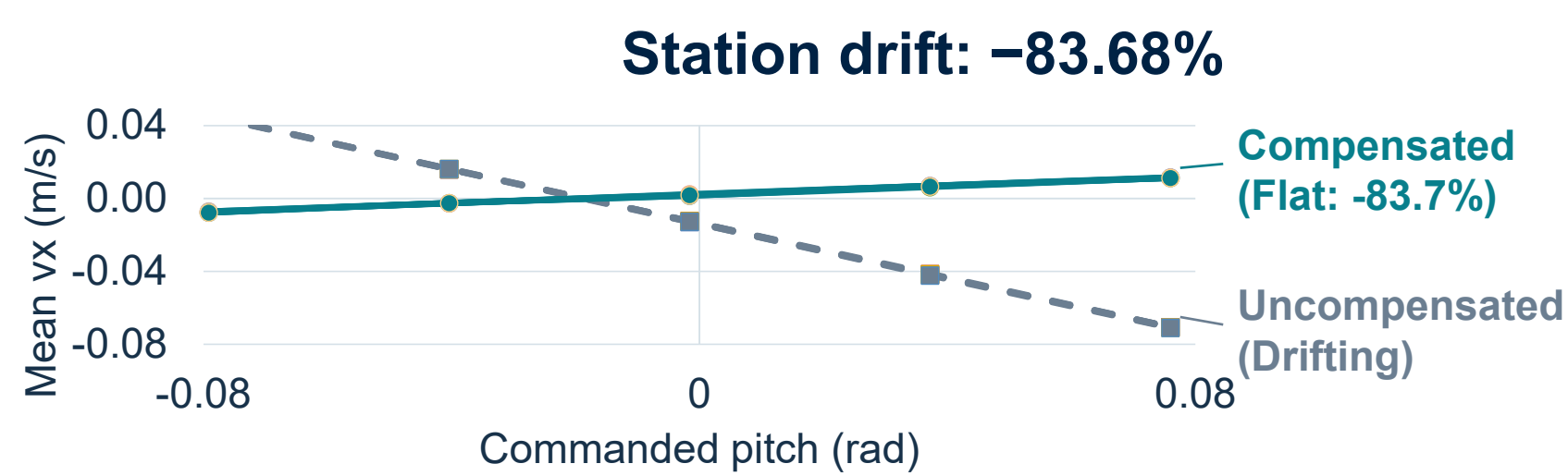
03 Operational boundary mapping

40 delay/noise conditions reveal where learned corrections become fragile.

Core Dilemma: Wheeled-legged locomotion exhibits severe wheel-leg dynamic coupling—changing leg posture alters balance equilibria, while wheels simultaneously govern locomotion and posture.

Our Paradigm: Analytical models resolve predictable steady-state drift; bounded residual RL improves high-impulse transient recovery.

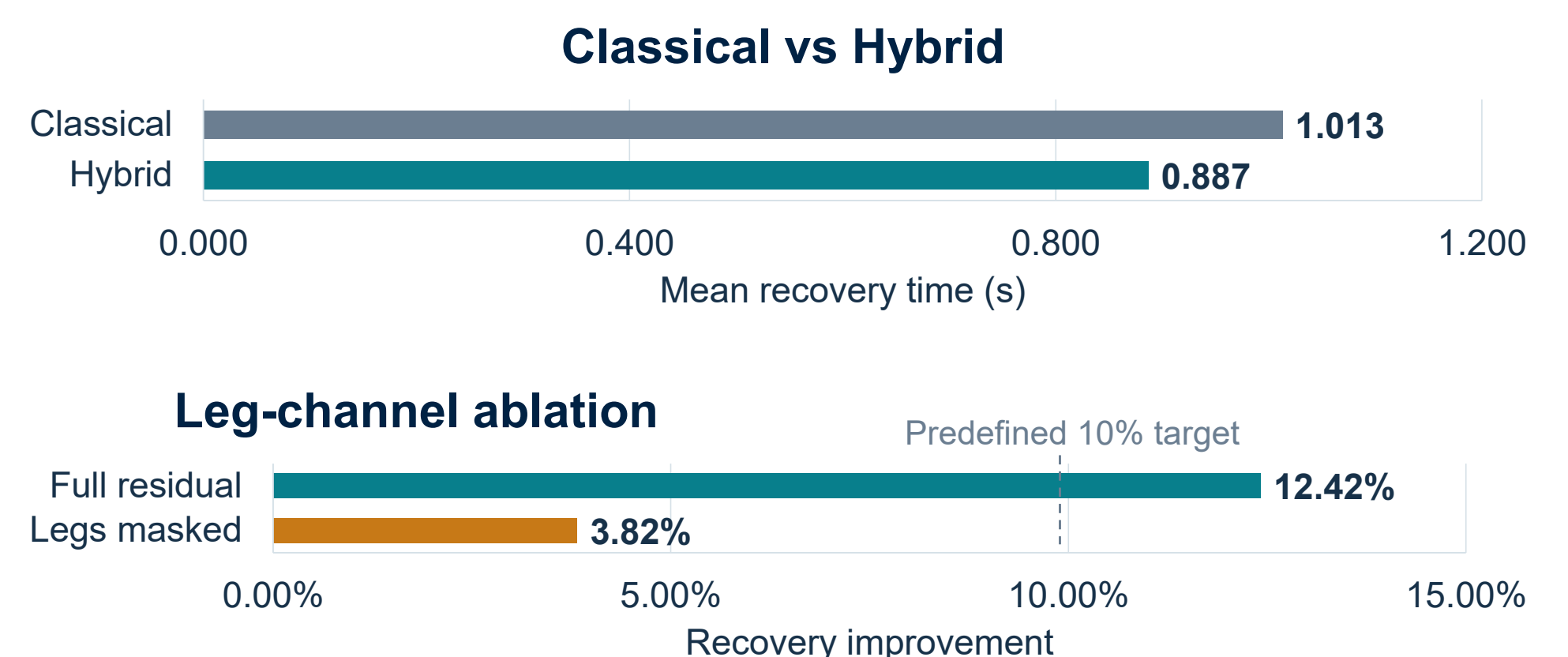
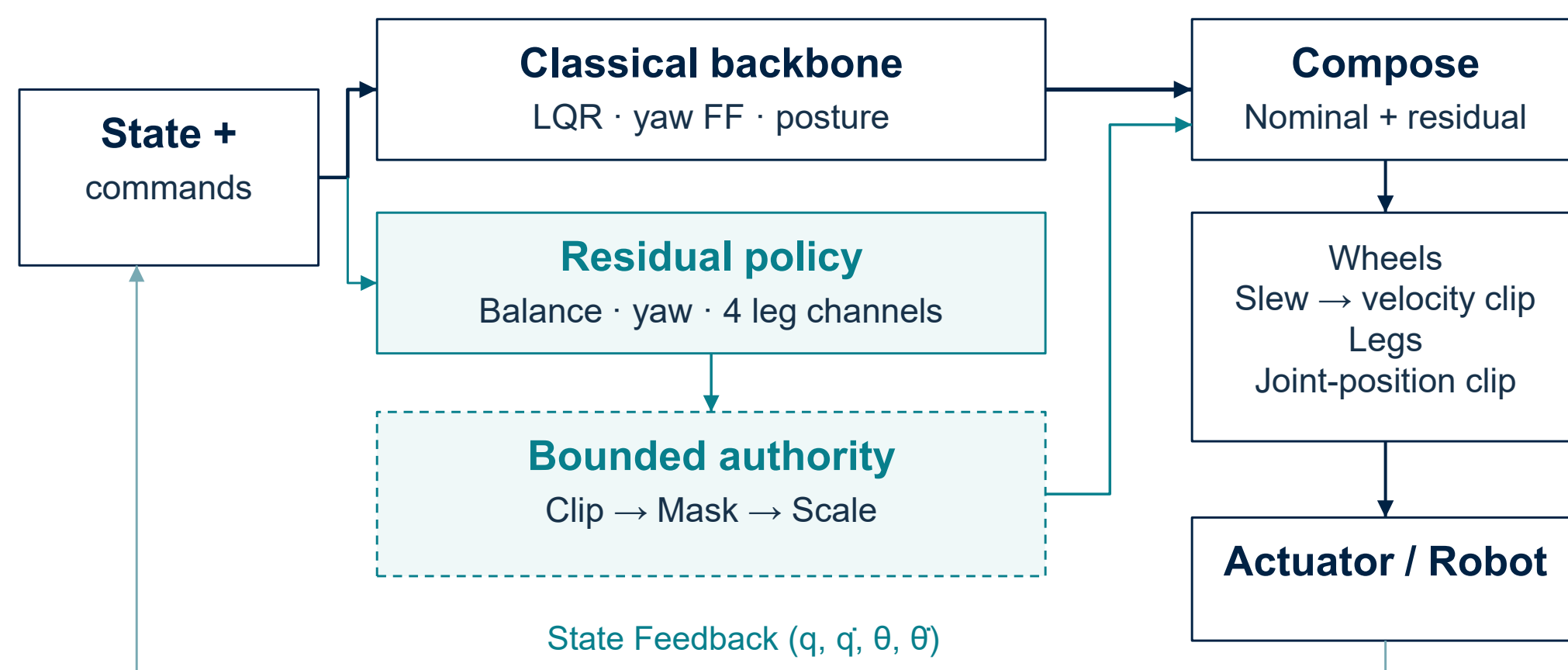
01 Nominal compensation & reference shaping



[Measured Trade-off] Command shaping suppresses velocity peaks by **94.44%** at the designed cost of settling time (**0.78 s** → **1.74 s**), reducing actuator shock before learning is applied.

Insight: Classical compensation resolves deterministic kinematics and drift, establishing a robust, non-strawman baseline.

02 Bounded residual learning & leg-channel ablation



Punchline: Full residual exceeds the 10% engineering target (**12.42%** faster recovery). Leg-channel ablation removes **~70% of the observed learning benefit**: the gain falls to **3.82%** with leg residuals masked, supporting wheel-leg coordination.

Simulation · one training seed · 128 events per arm · same checkpoint, evaluation-time masking · own matched baselines

03 Operational boundary: delay & noise

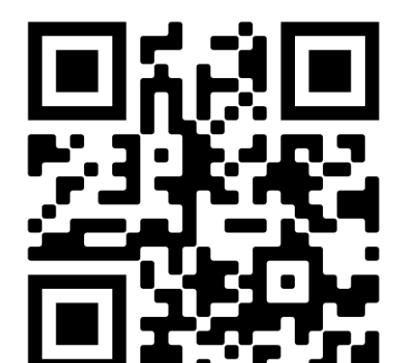
Recorded termination events · 40 conditions · shared scale 0–22

	Classical					Hybrid					
None	0	0	0	0	0	0	0	0	2	6	0
Encoder	0	0	0	0	0	0	0	0	0	3	
MEMS IMU	0	0	0	0	1	0	0	0	0	10	
MEMS IMU ×2	0	0	0	0	2	0	0	0	0	22	
	0	20	40	60	80	0	20	40	60	80	

Delay (ms)

Deployment Specification: Zero termination events through 40 ms support a provisional **≤40 ms control latency budget**. Higher delays expose residual fragility; validate this budget on hardware before physical deployment.

04 Simulation demo



Scan for Push-Recovery Demo

MuJoCo physics · Side-by-side comparison
Reproducible simulation workflow